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Introduction

- Sets, Relations and Functions is the beginning of the calculus
- Set Theory helps us to study the collection of objects and relations helps us to relate the objects in sets
- In real life we can relate function to an input output system.

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Contents

1 Definition and Description of Sets

2 Types of Sets

3 Operations on Sets

4 Cardinality based Problems

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Contents

1 **Definition and Description of Sets**

2 Types of Sets

3 Operations on Sets

4 Cardinality based Problems

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1. Definition and Description of Sets

A, B, C

$A = \{a, b, c\}$

$A = \{a, a, b, c\}$

$a \in A$

$\in$ belongs to

1. Definition and Description of Sets

Definition

A set is a collection of well defined objects.

NOTE

(a) Sets are generally denoted by capital letters like A, B, X, Y etc.
Elements of the set are denoted by small letters like a, b, x, y etc.

(b) If a is an element of the set A, then it's written as $a \in A$.

(c) Elements are not repeated in the set.

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1. Definition and Description of Sets

Description

Sets are described in two ways.

(i) Roster form

Elements are listed using commas and brackets

For example,

$\{a, e, i, o, u\}$ = A

(ii) Set Builder form

Elements are told by telling the property or rule they follow

For example,

$\{x : x \text{ is a vowel}\}$ = A

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Let's Solve



Let's Solve

6 : 1, 2, 3, 6

Ex. Match the following

(A) $\{1, 2, 3, 6\}$	(I) $\{x : x$ is a prime divisor of $6\}$
(B) $\{2, 3\}$	(II) $\{x : x \in \mathbb{N}$ and x is a divisor of $6\}$
(c) $\{A, C, E, H, I, M, S, T\}$	(III) $\{x : x$ is a letter of the word MATHEMATICS$\}$
	(IV) $\{x : x \in \mathbb{N}$ and $x < 7\}$
	(V) $\{x : x$ is a letter of the word ALCHEMIST$\}$

M, A, T, H, E, I, C, S

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Let's Solve

Ex. Match the following

(A) $\{1, 2, 3, 6\}$	(I) $\{x : x \text{ is a prime divisor of } 6\}$
(B) $\{2, 3\}$	(II) $\{x : x \in \mathbb{N} \text{ and } x \text{ is a divisor of } 6\}$
(c) $\{A, C, E, H, I, M, S, T\}$	(III) $\{x : x \text{ is a letter of the word MATHEMATICS}\}$
	(IV) $\{x : x \in \mathbb{N} \text{ and } x < 7\}$
	(V) $\{x : x \text{ is a letter of the word ALCHEMIST}\}$

Solution

$A \rightarrow II; B \rightarrow I; C \rightarrow III$

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Let's Solve

$A =$

Ex. Write the set $\{1, 2, 4, 8, 16, 32, 64\}$ in set-builder form.

$1, 2, 2^2, 2^3, 2^4, 2^5, 2^6$

$A = \{ 2^n ; n \in \text{whole numbers}, n \leq 6 \}$

$B = \{ 2^{n-1} ; n \in N ; n \leq 7 \}$



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Let's Solve

Ex. Write the set $\{1, 2, 4, 8, 16, 32, 64\}$ in set-builder form.

Solution

Given set is $A = \{1, 2, 4, 8, 16, 32, 64\}$

By observation, we can say

$A = \{2^{n-1} : n \leq 7 \text{ and } n \in \mathbb{N}\}$

Alternatively, we can say

$A = \{2^{n} : n \leq 6 \text{ and } n \in W\}$

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Let's Solve

Ex. Write the set $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}\right\}$ in set-builder form.

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Let's Solve

Ex. Write the set $\left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}\right\}$ in set-builder form.

Solution

Given set is $A = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}\right\}$

By observation, we can say

$$A = \left\{\frac{x}{x+1} : x \leq 4 \text{ and } x \in \mathbb{N}\right\}$$

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Quiz.



Quiz.

Chat Off

Q. What is the set builder form of the set $\{1, 3, 7, 15, 31\}$?

A $\{2^{n-1} : n < 5 \text{ and } n \in \mathbb{N}\}$

B $\{2n-1 : n \leq 5 \text{ and } n \in \mathbb{N}\}$

C $\{2^n + 1 : n \leq 6 \text{ and } n \in \mathbb{N}\}$

D $\{2^n - 1 : n < 6 \text{ and } n \in \mathbb{N}\}$

$2^n - 1 : n \leq 5, n \in \mathbb{N}$

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Quiz.

Chat On

Q. What is the set builder form of the set $\{1, 3, 7, 15, 31\}$?

A $\{2^{n-1} : n < 5 \text{ and } n \in \mathbb{N}\}$

B $\{2n-1 : n \leq 5 \text{ and } n \in \mathbb{N}\}$

C $\{2^n + 1 : n \leq 6 \text{ and } n \in \mathbb{N}\}$

D $\{2^n - 1 : n < 6 \text{ and } n \in \mathbb{N}\}$



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Quiz.

Q. What is the set builder form of the set $\{1, 3, 7, 15, 31\}$?

Solution

Given set is $A = \{1, 3, 7, 15, 31\}$

By observation, we can say

$A = \{2^n - 1 : n \leq 5 \text{ and } n \in \mathbb{N}\}$ or $A = \{2^n - 1 : n < 6 \text{ and } n \in \mathbb{N}\}$



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Contents

1 Definition and Description of Sets

2 **Types of Sets**

3 Operations on Sets

4 Cardinality based Problems

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2. Types of Sets

2. Types of Sets

Null set or Empty set or Void set

A set which does not contain any element is called Empty set or Null set or Void set. It's denoted by ϕ or $\{\}$.

Ex The sets $\{x : x^2 + 1 = 0 \text{ and } x \in \mathbb{R}\}$ and $\{x : 2x - 3 = 0 \text{ and } x \in \mathbb{Z}\}$ are null sets.

SINGLETON SET

$$A = \{1\}$$

$$A = \{a\}$$

$$A = \{\{a\}\}$$

2. Types of Sets

Singleton Set

A set containing single element is called singleton set.

Ex $\{1\},\{0\},\{\phi\},\{\{2,3\}\}$ are singleton sets.

NOTE

$\{\{2,3\}\}$ has one element, that is $\{2,3\}$

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Quiz.



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Quiz.

Chat Off

Q. Which of the following is a singleton set?

A {2, 3}

B { }

C {M}

D none



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Quiz.

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Q. Which of the following is a singleton set?

A {2, 3}

B { }

C {M}

D none



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Quiz.

Q. Which of the following is a singleton set?

Solution

{2, 3} contains 2 elements, namely, 2 and 3

{ } contains no elements

{M} contains 1 element, namely, M



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FINITE SET

Ex:

2. Types of Sets

Finite Set

A set which is either empty or has finite number of elements is called a finite set.

$A = \{1, 2, 3, 6, 7\}$

$n(A) = 5$

7

Remark

Number of elements in a finite set A is called the order or the cardinality of the set A, generally denoted by $o(A)$ or $n(A)$.

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Quiz.



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Quiz.

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Q. If $A = \{1, 2, \{3, 4\}\}$, then $n(A)$ _____.

A 2

B 3

C 4

D None



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Quiz.

Chat On

Q. If $A = \{1, 2, \{3, 4\}\}$, then $n(A)$ _____.

A 2

B 3

C 4

D None



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Quiz.

Q. If $A=\{1,2,\{3,4\}\}$, then $n(A)$ _____.

Solution

Given $A=\{1,2,\{3,4\}\}$

Here, $\{3,4\}$ is an element of A

So, $n(A)=3$



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2. Types of Sets

Infinite Set

A set having infinite number of elements is called infinite set.

Ex $\{1,2,3,4,...\},\{...,-2,-1,0,1,2,...\}$ etc. are infinite sets.

2. Types of Sets

Equal Sets

Two sets are said to be equal if they have exactly the same elements.

Ex If $A=\{0,1,2\}, B=\{2,1,0\}, C=\{0,1,1,2\}$, then $A=B=C$.

Equivalent Sets

Two finite sets are said to be equivalent if their cardinalities are equal.

A B

Ex $\{1,2,3\}$ is equivalent to {dog, cat, parrot}.

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2. Types of Sets

SUBSET

$\subseteq$

$A \subset B$

$B = \{1, 2, 3, 4\}$

$A = \{1, 2\}$

$A \subset B$

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2. Types of Sets

Definition

Set A is said to be a subset of set B if every element of A is also an element of B. It is generally denoted as $A \subset B$.

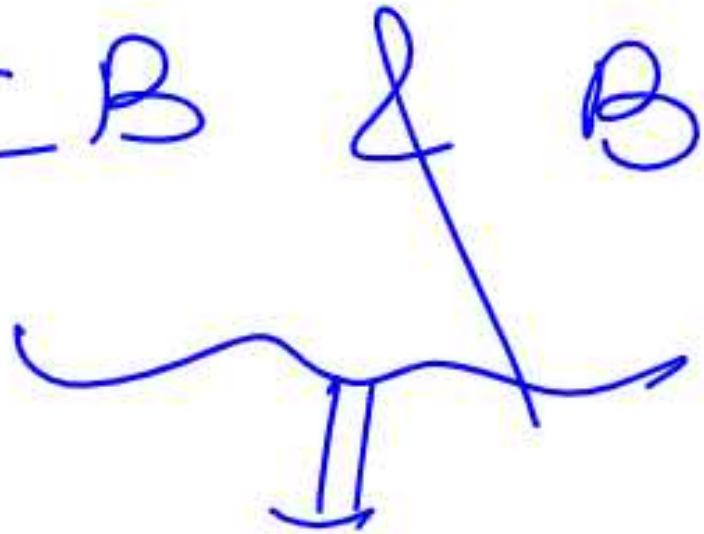
$A = \{1, 2, 3\}$

$= \{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{3, 1\}, \{1, 2, 3\}$

Total no. of subsets $= 2^{n(A)}$

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$A \subset B \;\&\; B \subset A$



$$A = B$$

2. Types of Sets

Definition

Set A is said to be a subset of set B if every element of A is also an element of B. It is generally denoted as $A \subset B$.

Ex If $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$, then $A \subset B$.

NOTE

(a) Every set is a subset of itself.

(b) ϕ is a subset of all sets.

(c) $A \subset B$ and $B \subset A \Rightarrow A = B$.

2. Types of Sets

Observation.

If $A = \{a, b, \{1,2,3\}\}$, then try to observe that

(i) $\{a,b\} \subset A$ → T

(ii) $\{\{1,2,3\}, b\} \subset A$ → T

(iii) $\phi \subset A$ → T

2. Types of Sets

POWER SET $P(A)$

$A = \{1, 2, 3\}$

$P(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{3,1\}, \{1,2,3\}\}$

2. Types of Sets

Definition

The collection of all the subsets of a set A is called the power set of A, denoted by P(A).

Let's Solve



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Let's Solve

Ex. Write down the power sets of the following sets.

(a) $A = \{1, 2\}$

(b) $A = \{1, 2, 3\}$

$P(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{2, 3\}, \{1, 2\}\}$

2^n

$n(P(A)) =$ no. of subsets of A

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Let's Solve

Ex. Write down the power sets of the following sets.

(a) $A = \{1, 2\}$

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Let's Solve

Ex. Write down the power sets of the following sets.

(a) $A = \{1, 2\}$

Solution

Given $A = \{1, 2\}$

Power set of $A = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$

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Let's Solve

Ex. Write down the power sets of the following sets.

(b) $A = \{1, 2, 3\}$

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Let's Solve

Ex. Write down the power sets of the following sets.

(b) $A = \{1, 2, 3\}$

Solution

Given $A = \{1, 2, 3\}$

$P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}\}$

2. Types of Sets

Result

If $n(A) = p$, then the number of subsets of A is 2^p.

Clearly, same will be $n(P(A))$.

Quiz.



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Quiz.

Chat Off

Q. If $n(A)=3$, then $n(P(A))=$ _____.

8

A 6 B 3 C 8 D None

Total no. of subsets $= 2^3 = 8$



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Quiz.

Chat On

Q. If $n(A) = 3$, then $n(P(A)) =$ _____.

A 6

B 3

C 8

D None



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Quiz.

Q. If $n(A)=3$, then $n(P(A))=$ _____.

Solution

Given, $n(A)=3$

$\therefore n(P(A))=2^{n(A)}=2^3=8$



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Let's Solve



Let's Solve

Ex. Two finite sets have p and q number of elements Respectively. The number of elements in the power set of the first set is 56 more than the number of elements in the power set of the second set. Find p and q.

$n(A) = p$; $n(B) = q$

$$n(P(A)) = n(P(B)) + 56$$

$$2^p = 2^q + 56$$

$$2^p - 2^q = 56$$



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$$2^p - 2^q = 56$$

$$2^q \left[2^{p-q} - 1 \right] = 56$$

$$= 8 \times 7$$

$$= 2^3 \times (8-1)$$

$$2^q \left[2^{p-q} - 1 \right] = 2^3 \times (2^3 - 1)$$

$$q = 3$$

$$p - q = 3$$

$$p = 6$$

Let's Solve

Ex. Two finite sets have p and q number of elements Respectively. The number of elements in the power set of the first set is 56 more than the number of elements in the power set of the second set. Find p and q.

Solution

Here, let the two sets be A and B,

such that $n(A) = p$ and $n(B) = q$.

Number of elements in power set of $A = 2^p$

Number of elements in power set of $B = 2^q$

Now, according to question, $2^p - 2^q = 56,$,

that is $2^q\left(2^{p-q} - 1\right) = 2^3\left(2^3 - 1\right)$

$\therefore q = 3, q = 6$

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Quiz.



Quiz.

Chat Off

Q. Two finite sets have m and $n(m > n)$ elements. The number of subsets of the first set is 112 more than that of second set. The value of mn is

A 26 B 28 C 32 D 36

$$2^m = 112 + 2^n$$

$$2^m - 2^n = 112$$

$$2^n[2^{m-n} - 1] = 112$$

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$$2^n\left[2^{m-n}-1\right]=112=56\times 2$$

$$=7\times 8\times 2$$

$$=16\times 7$$

$$=2^4\left[2^3-1\right]$$

$$n=4$$

$$m-n=3$$

$$m=4+3=7$$

Quiz.

Chat On

Q. Two finite sets have m and $n(m > n)$ elements. The number of subsets of the first set is 112 more than that of second set. The value of mn is

A 26

B 28

C 32

D 36

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Quiz.

Q. Two finite sets have m and $n(m > n)$ elements. The number of subsets of the first set is 112 more than that of second set. The value of mn is

Solution

According to the question $2^m - 2^n = 112$

$2^n\left(2^{m\ \ n} - 1\right) = 2^4 \times 7$

Comparing both sides

$2^n = 2^4$ and $2^{m-n} - 1 = 7$

$\Rightarrow n = 4$ and $2^{m-n} = 8$

$\Rightarrow n = 4$ and $m - n = 3$

$\Rightarrow n = 4$ and $m = 7$

$\Rightarrow mn = 28$

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Contents

1 Definition and Description of Sets

2 Types of Sets

3 **Operations on Sets**

4 Cardinality based Problems

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3. Operations on Sets

To understand operations on sets, first we need to know the term "universal set".

3. Operations on Sets

Universal Set

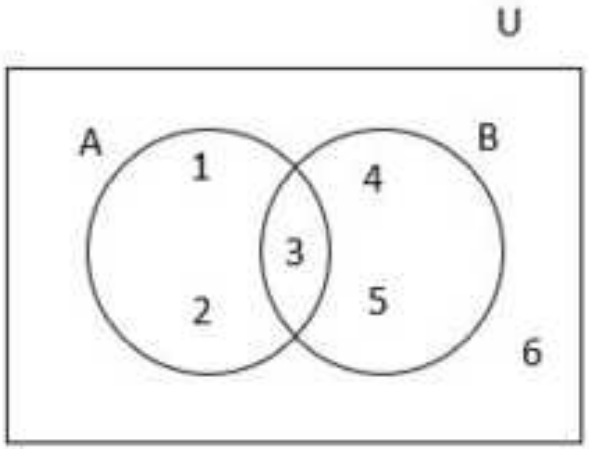
If we talk about any particular context in sets then the set which is superset of all possible sets in the given context is called universal set.
(Generally, it's given in the question).

Venn Diagram

Ex $A = \{1,2,3\}$

$B = \{3,4,5\}$

$U = \{1,2,3,4,5,6\}$



3. Operations on Sets

Following are the operations on sets that we are now going to learn.

(a) Union of two Sets

(b) Intersection of two Sets

(c) Difference of two Sets

(d) Symmetric Difference of two Sets

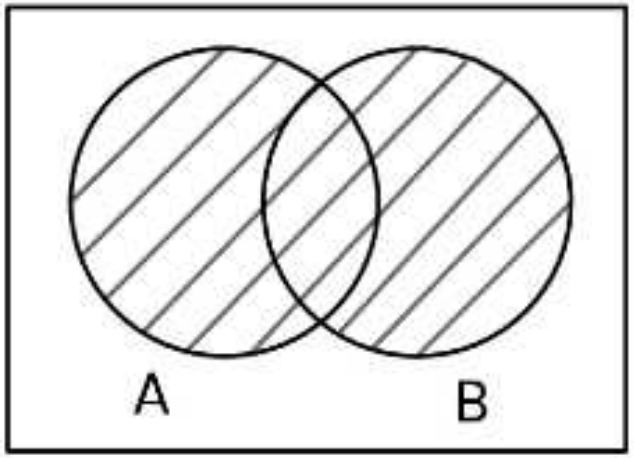
(e) Complement of a Set

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3. Operations on Sets

Union of two Sets

$A \cup B = \{x : x \in A \text{ or } x \in B\}$



$A = \{1, 2, 3, 4\}$

$B = \{4, 5, 6\}$

$A \cup B = \{1, 2, 3, 4, 5, 6\}$

Ex $\{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}$

3. Operations on Sets

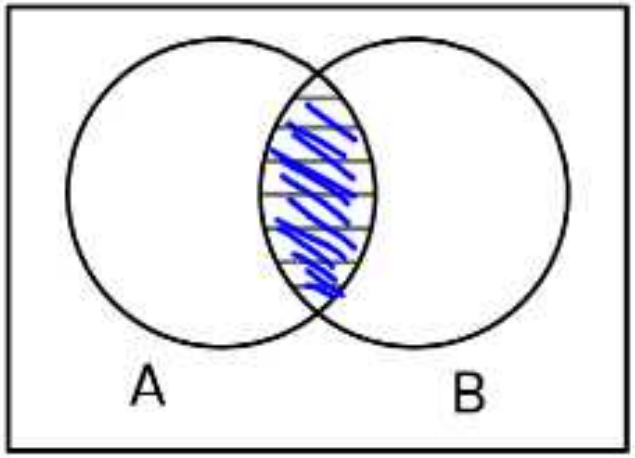
Intersection of two Sets

$A \cap B = \{x : x \in A \text{ and } x \in B \text{ simultaneously}\}$

$A = \{1, 2, 3, 4, 5\}$

$B = \{4, 5, 6, 7\}$

$A \cap B = \{4, 5\}$

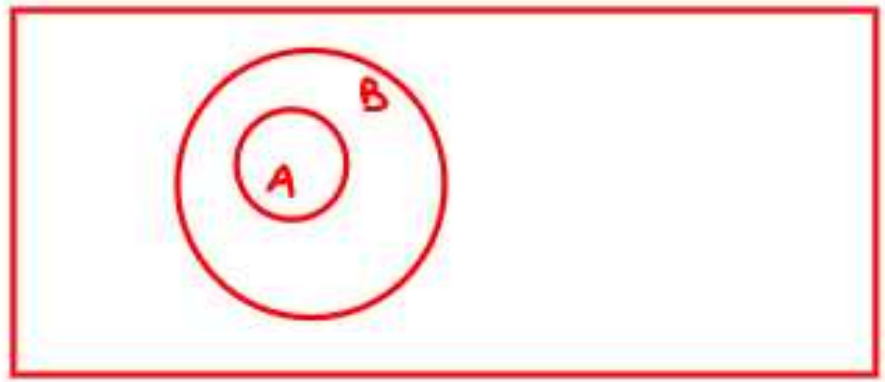


Ex $\{1,2,3\} \cap \{2,3,4\} = \{2,3\}$

i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ ✓

ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ ✓

iii) $A \subset B$



$A \cup B = B$

$A \cap B = A$

$A \cap B = \phi$

A & B => disjoint sets

3. Operations on Sets

NOTE

(a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C); A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) If $A \subset B$, then

(i) $A \cup B = B$

(ii) $A \cap B = A$

(c) If $A \cap B = \phi$, then A and B are called disjoint sets.

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Difference Of Two Sets

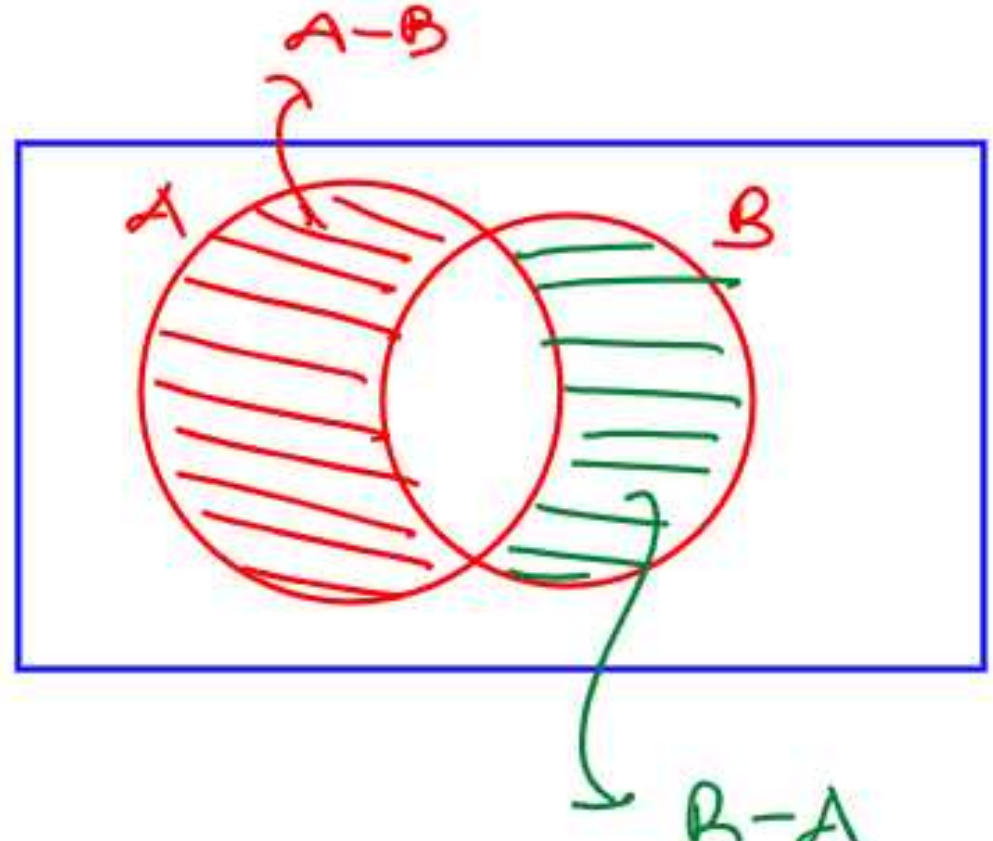
A − B

$A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

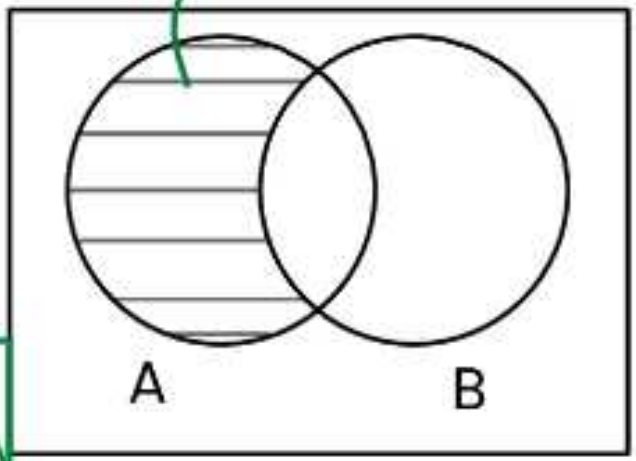
$A - B = \{1, 2\}$

$B - A = \{5, 6\}$



3. Operations on Sets

A - B



Difference of two Sets

$A - B (\text{or}) A / B = \{x : x \in A \text{ and } x \notin B\}$

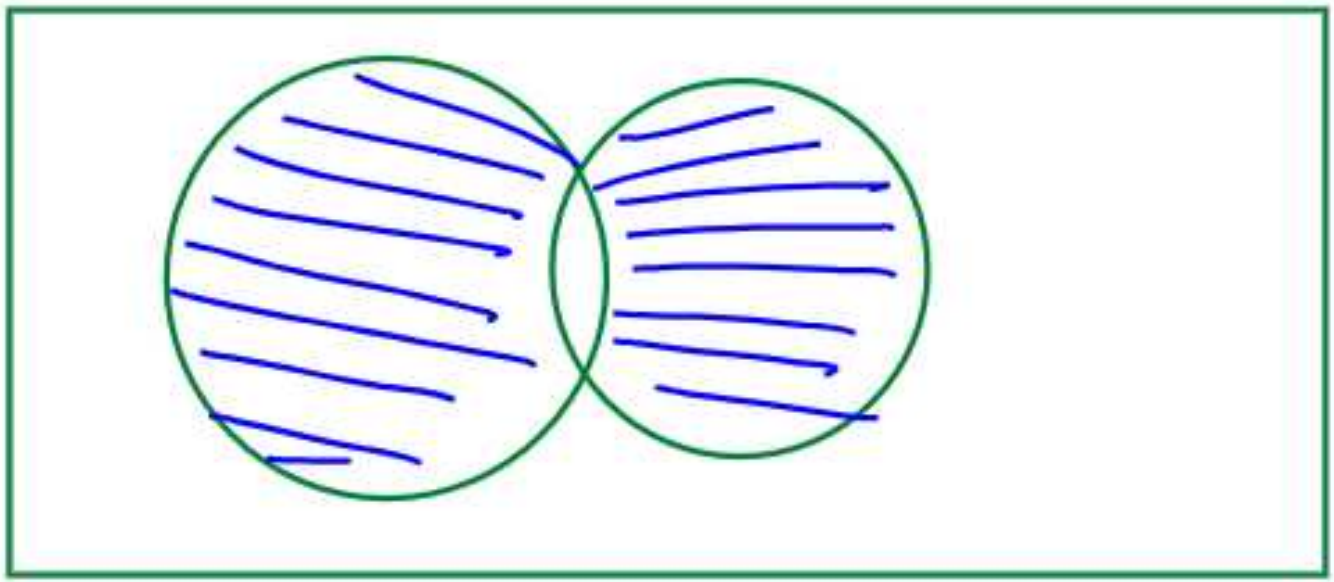
$n(A/B) = n(A) - n(A \cap B)$

Ex $\{1,2,3\} - \{2,3,4\} = \{1\}$

Ex $\{1,2,3\} / \{4,5\} = \{1,2,3\}$

SYMMETRIC DIFFERENCE b/w 2 SETS

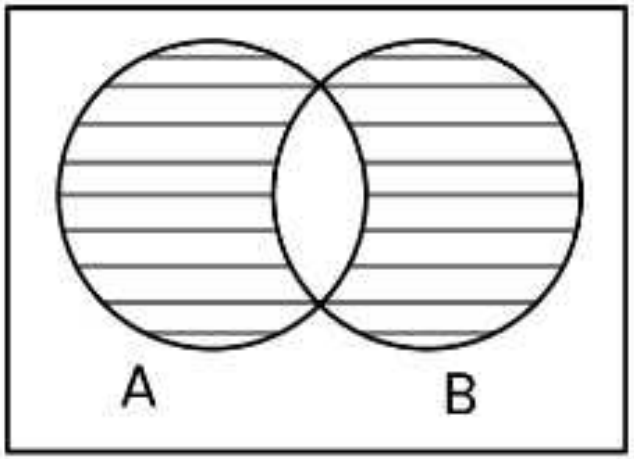
$A \triangle B \equiv A/B \cup B/A$



3. Operations on Sets

Symmetric Difference of two Sets

$$A \Delta B = (A / B) \cup (B / A)$$

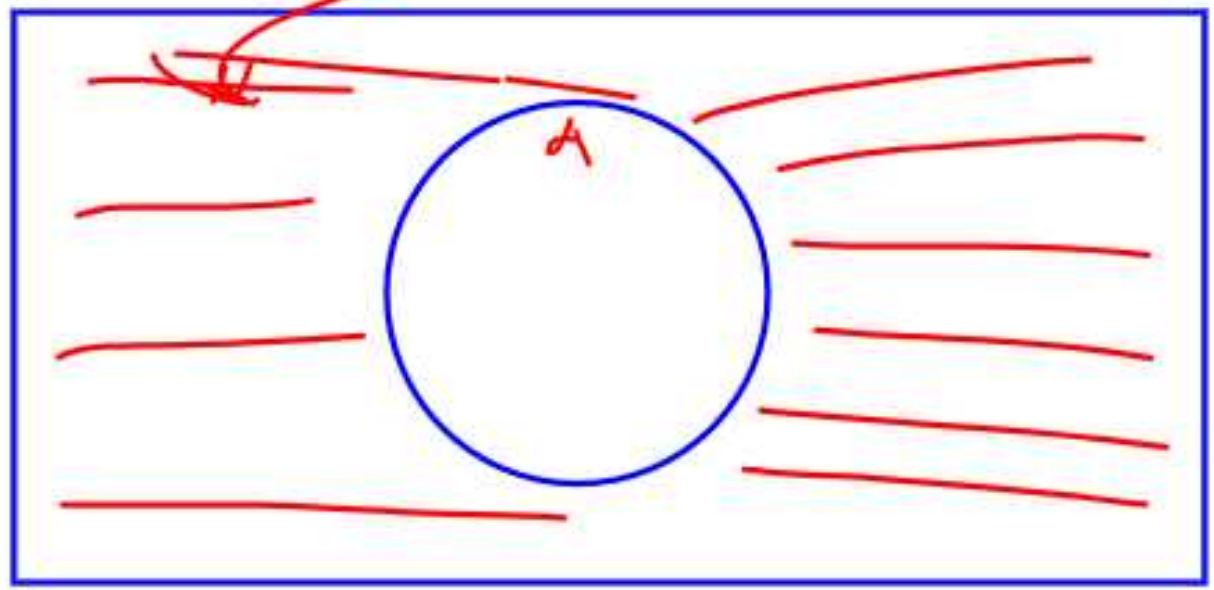


Observation.

$$A / B = A \cap B^{c}$$

COMPLEMENT SET

A^c / A' / $\bar{A}$

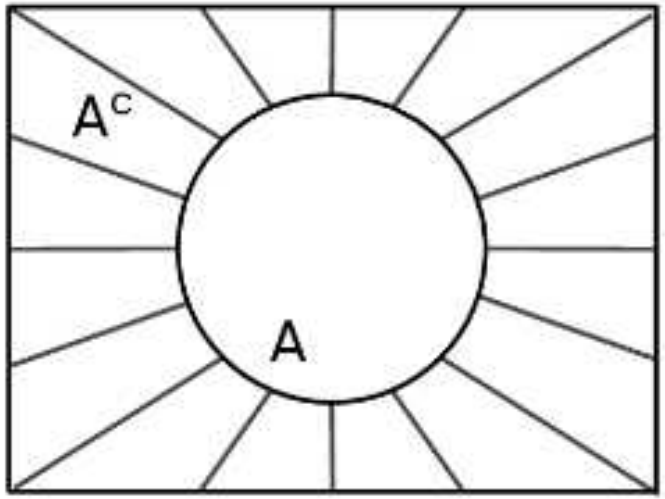


$$n(A') = n(S) - n(A)$$

3. Operations on Sets

Complement of a Set

A^c or $A' = \{x : x \notin A \text{ and } x \in U\}$



Ex If $U = \{1,2,3,4,5,6\}$ then complement of $\{1,2,3,4\}$ is $\{5,6\}$

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$A, \quad A^c$

i) $(A^c)^c = A$

ii) $A \cup A^c = S$

iii) $A \cap A^c = \phi$

De Morgan's Law

i) $n(\overline{A \cup B}) = n(\bar{A} \cap \bar{B})$

ii) $n(\overline{A \cap B}) = n(\bar{A} \cup \bar{B})$

3. Operations on Sets

De-Morgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Try to observe these using Venn Diagrams

Quiz.



Quiz.

Q. If $A=\{1,2,3\}, B=\{1,2,4\}$ and $U=\{1,2,3,4,5\}$, then which of the following is false?

A $A \cup B=\{1,2,3,4\}$

B $A \cap B=\{1,2\}$

C $A / B=\{1,4\}$

$\{3\}$

D $B^c / A=\{5\}$

$B^c=\{3,5\}$

$B^c - A=\{5\}$

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Quiz.

Chat On

Q. If $A=\{1,2,3\}, B=\{1,2,4\}$ and $U=\{1,2,3,4,5\}$, then which of the following is false?

A $A \cup B=\{1,2,3,4\}$

B $A \cap B=\{1,2\}$

C $A / B=\{1,4\}$

D $B^c / A=\{5\}$



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Quiz.

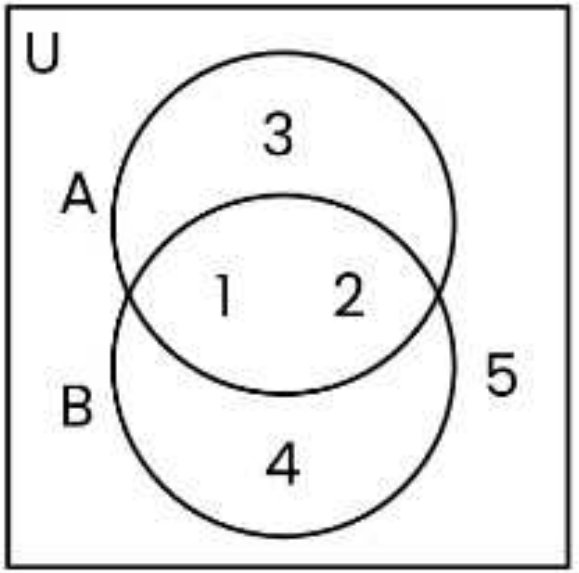
Q. If $A=\{1,2,3\}, B=\{1,2,4\}$ and $U=\{1,2,3,4,5\}$, then which of the following is false?

Solution

$A=\{1,2,3\}\quad B=\{1,2,4\}\quad \cup=\{1,2,3,4,5\}$

$A\cup B-\{1,2,3,4\}\quad A\cap B-\{2\}$

$A/B=\{3\}\quad B^{c}/C=\{5\}$



$A' / A^c / \bar{A}$ $\Rightarrow$ Complement of A

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) -$$

$$n(A \cap B) - n(B \cap C) - n(C \cap A)$$

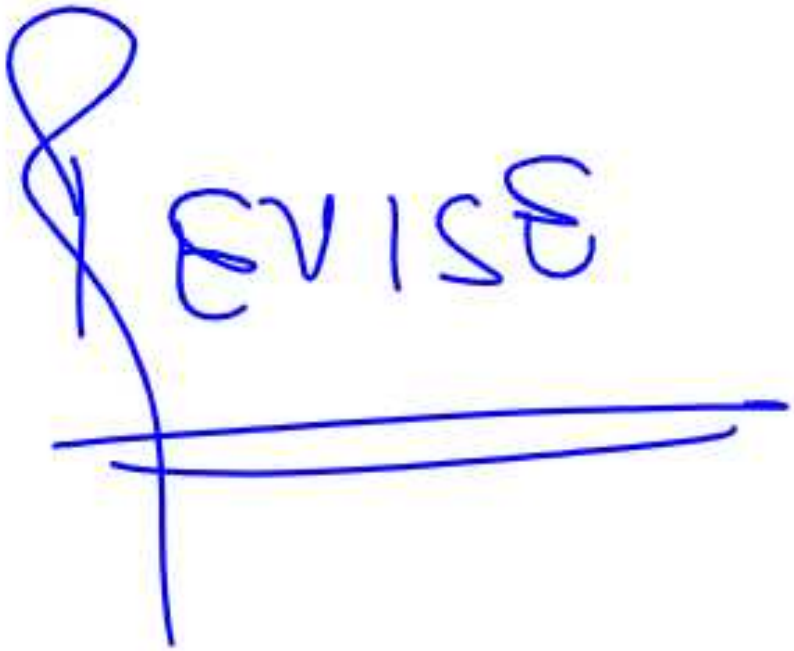
$$+ n(A \cap B \cap C)$$

Summary.



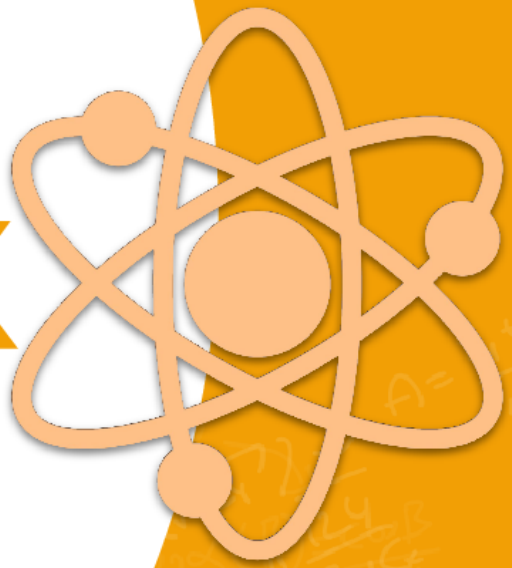
Summary.

1. Definition and Description of Sets
2. Types of Sets
3. Operations on Sets



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